

Dhirubhai Ambani University | 2025

Time Series Insights on Transport and Fuel Prices

Implication on Transportation Industry

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Abstract

We analyzed time series data of flight and railway ticket prices, along with kerosene and diesel fuel prices. Key steps included trend and seasonality removal, stationarity testing (ADF, ACF), and cross-correlation analysis. We also studied distribution fitting of residuals (Normal vs. Cauchy). Strong seasonal trends and meaningful correlations were found, especially during the COVID-19 period — all without forecasting.

Objectives

01

Analyze and decompose time series data of four monthly datasets (Flights, Kerosene, Railway, Diesel)

02

Identify trends, seasonality, and outliers

03

Perform stationarity tests and Distribution fitting

04

Evaluate cross-correlation between related pairs (Flights/Kerosene, Railways/Diesel)

Methodology

01

Data Import & Inspection

02

Trend Detection and Detrending

03

Seasonal Decomposition

04

Residual Analysis

05

SARIMA Modeling

06

Distribution Fitting

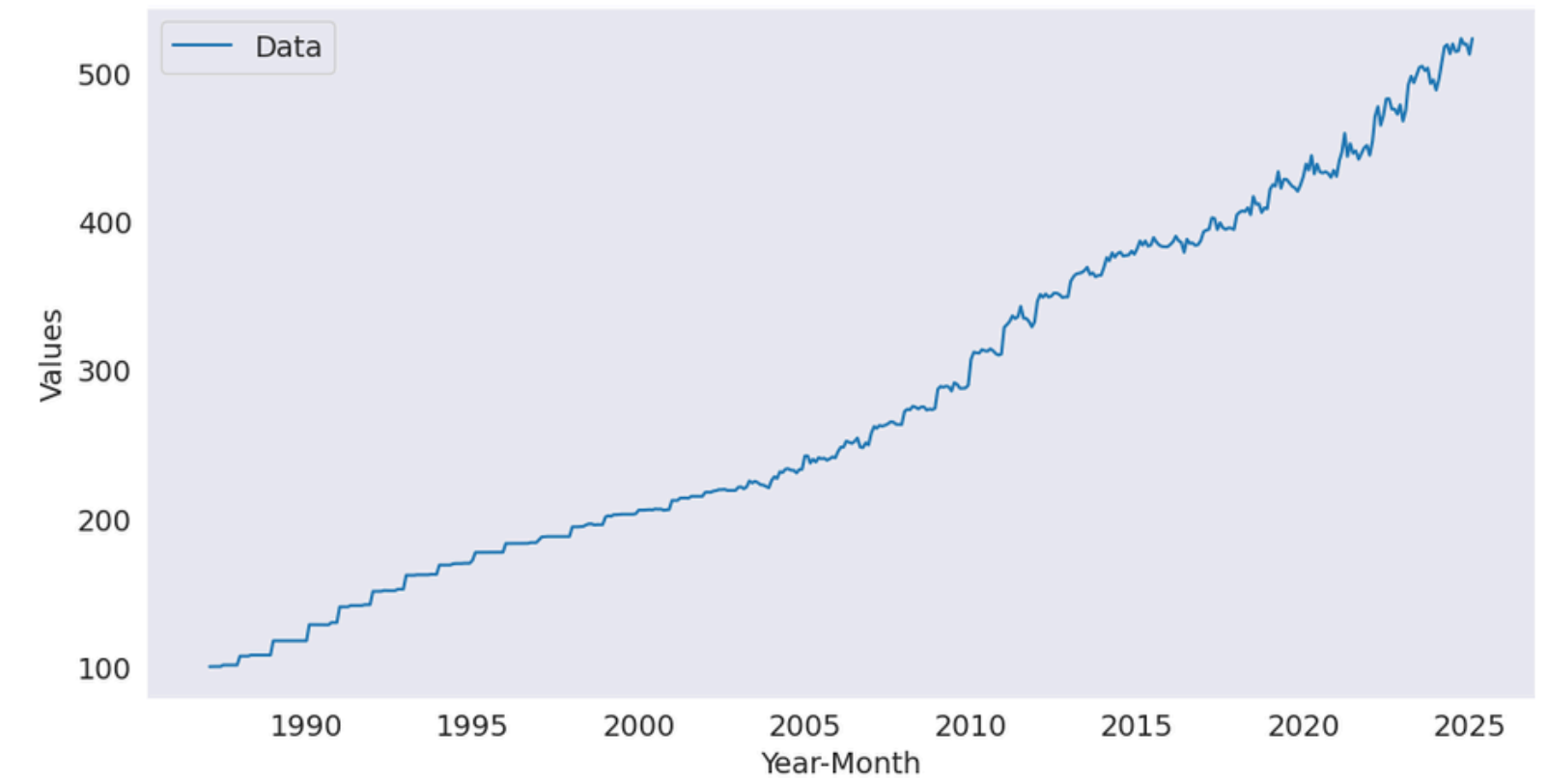
Data Overview

Time Series Data



Diesel Prices

Time Series Data



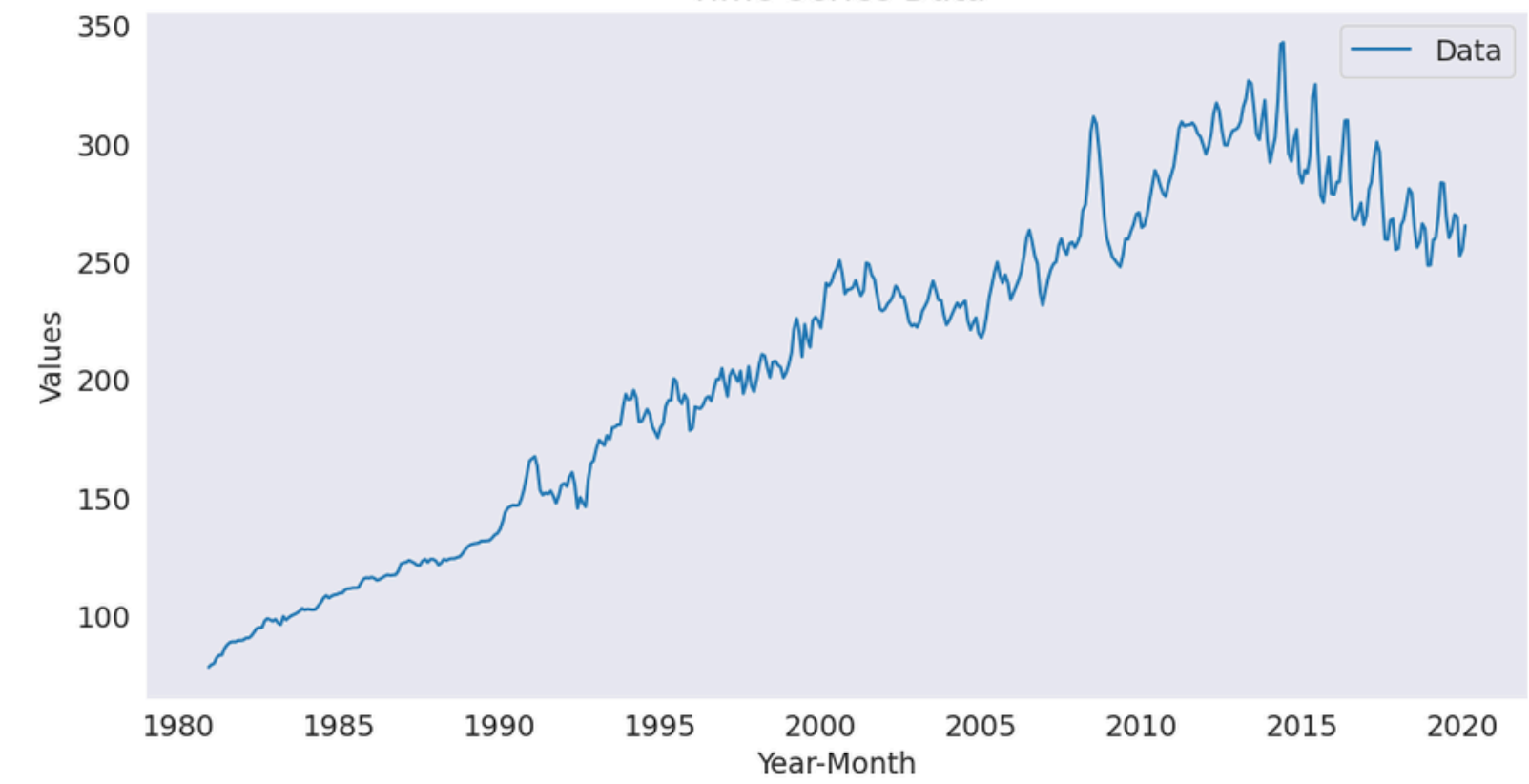
Railway Prices

Time Series Data



Kerosene Prices

Time Series Data

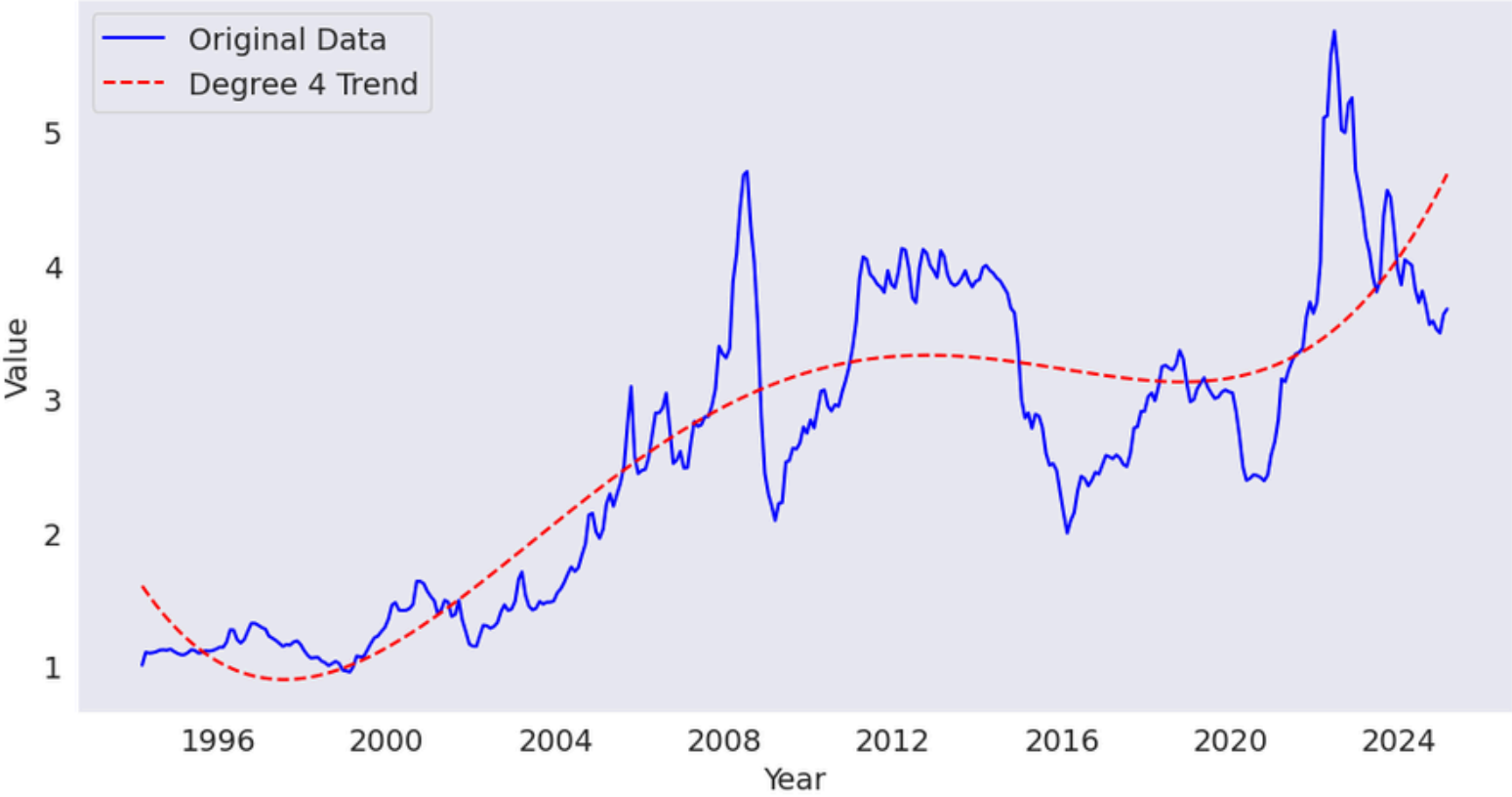


Flights Prices

Trends & Seasonality

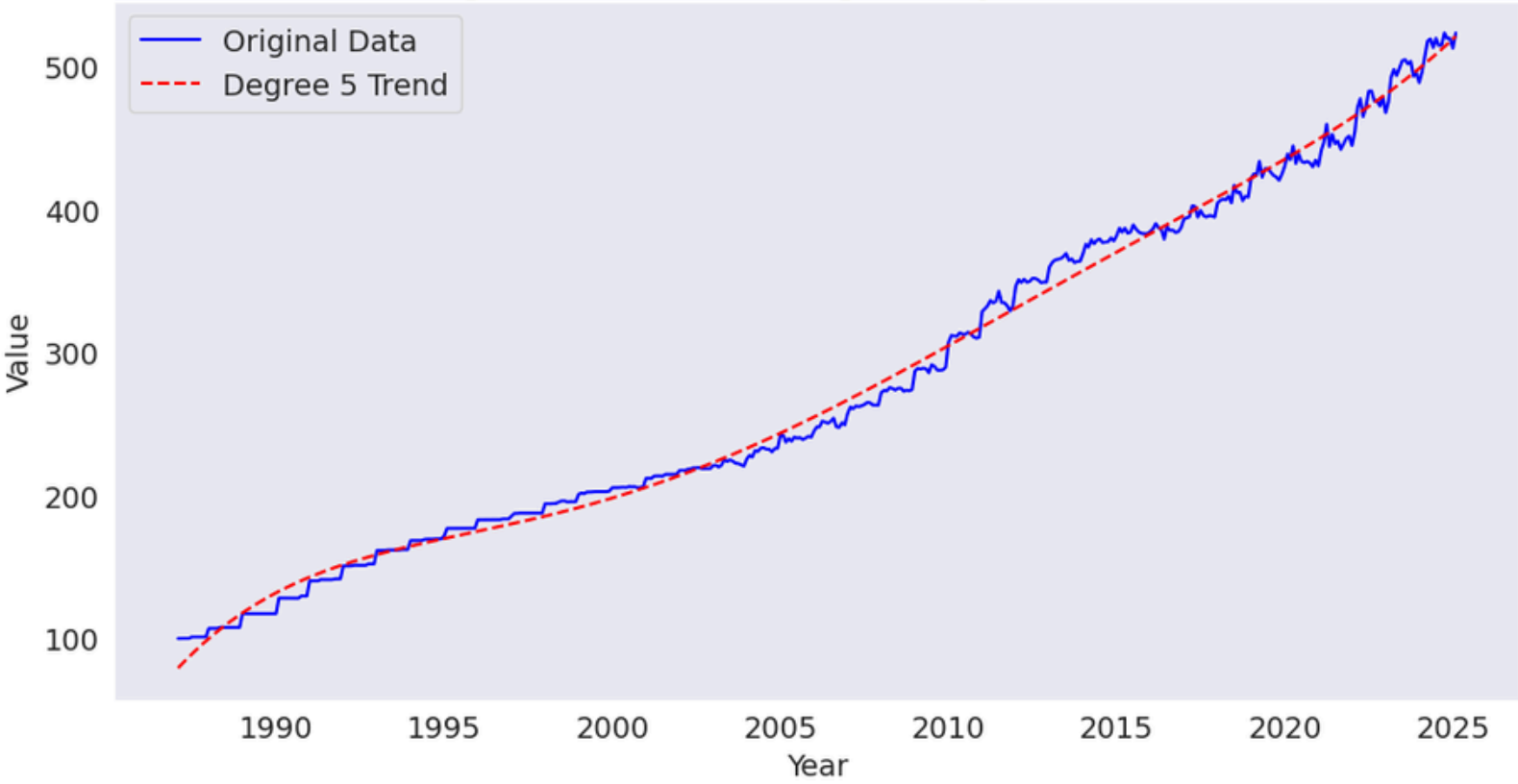
Diesel Prices

Original Data vs. 4th-Degree Polynomial Trend

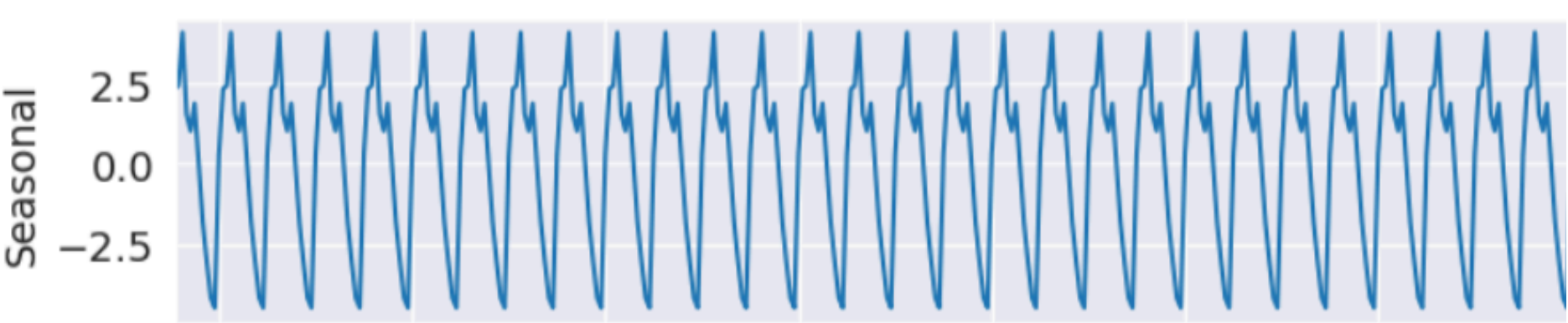
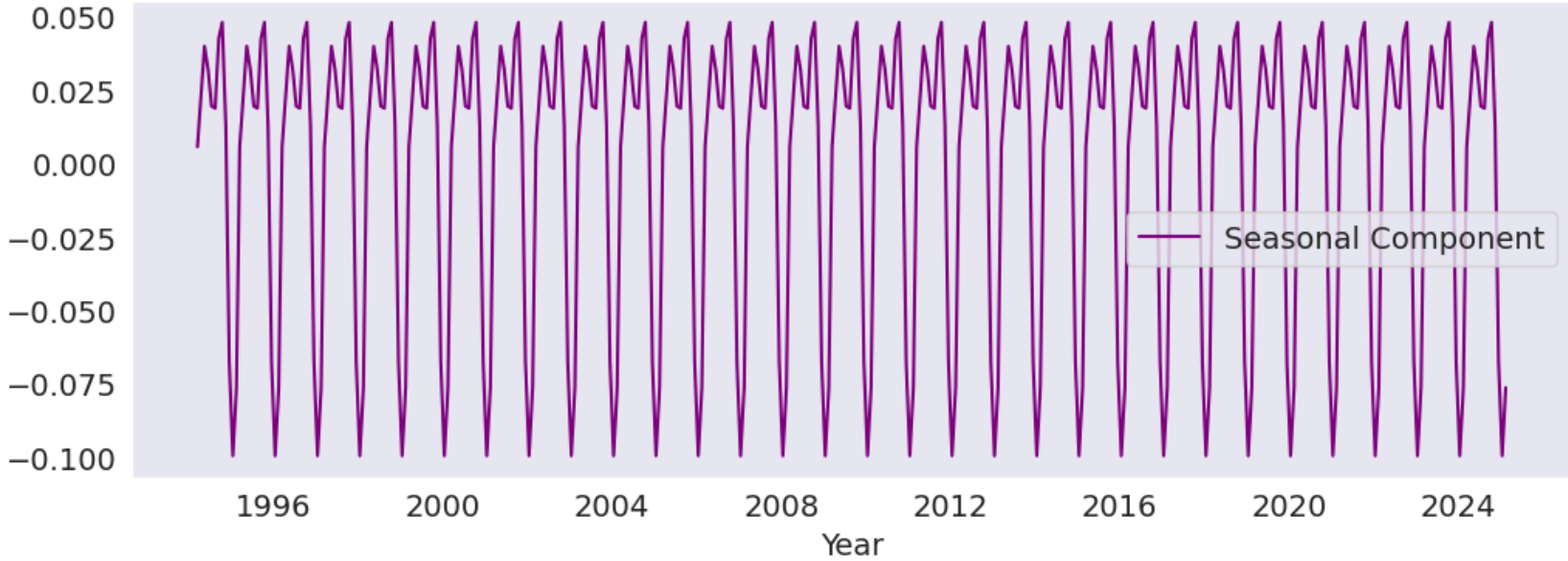


Railway Prices

Original Data vs. 5th-Degree Polynomial Trend



Extracted Seasonal Component



Trends & Seasonality

Diesel Prices

Trend: The data displays a long-term increasing pattern with significant fluctuations. This trend was captured using a 4th-degree polynomial, which effectively models the non-linear growth and turning points across years.

Seasonality: The seasonal component reveals recurring yearly patterns, likely influenced by global oil price cycles and policy-driven changes in fuel pricing. These effects cause regular short-term variations superimposed on the overall trend.

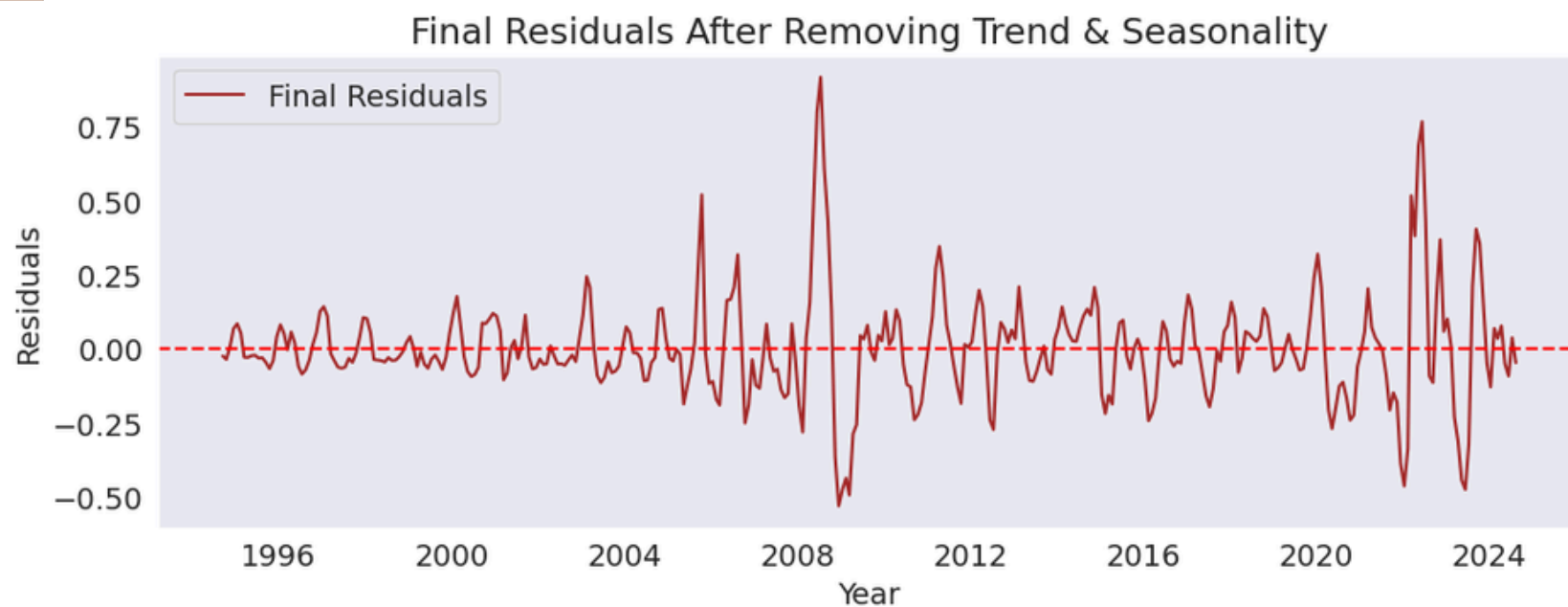
Railway Prices

Trend: The railway ticket prices show a gradually increasing pattern with notable non-linear shifts over time. A 5th-degree polynomial trend line was used to capture these complex variations, including periods of both sharp rises and dips.

Seasonality: Seasonal patterns appear consistently across years, potentially driven by holiday travel peaks, railway budget announcements, and periodic fare revisions, resulting in cyclical behavior in the data.

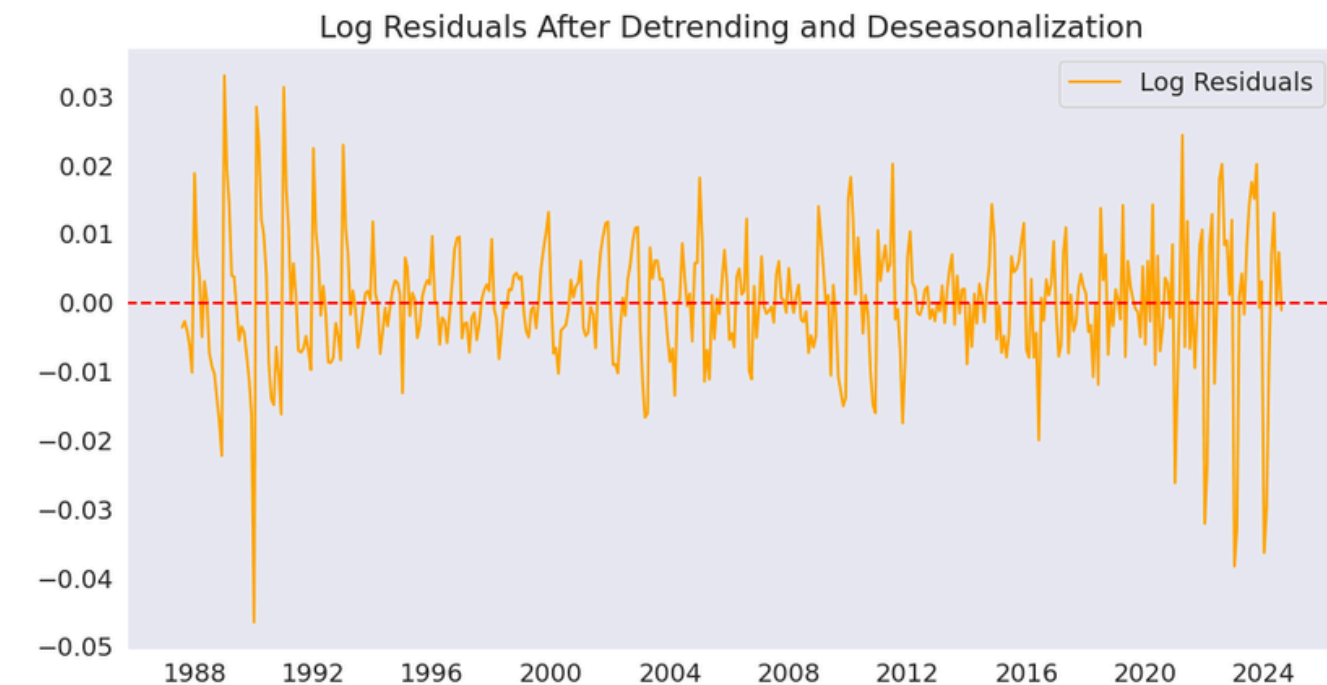
Residuals Analysis

Diesel Prices



- **Residuals:** Noticeable fluctuations appear in the periods 2007–2009 and 2022–2023, reflecting significant market shifts.
- **Key Abnormalities:** 2007–2009 (Recession): Price spike followed by a sharp fall due to low demand during economic downturn.
- 2022 (Russia-Ukraine War): Price volatility driven by geopolitical tensions and supply issues.

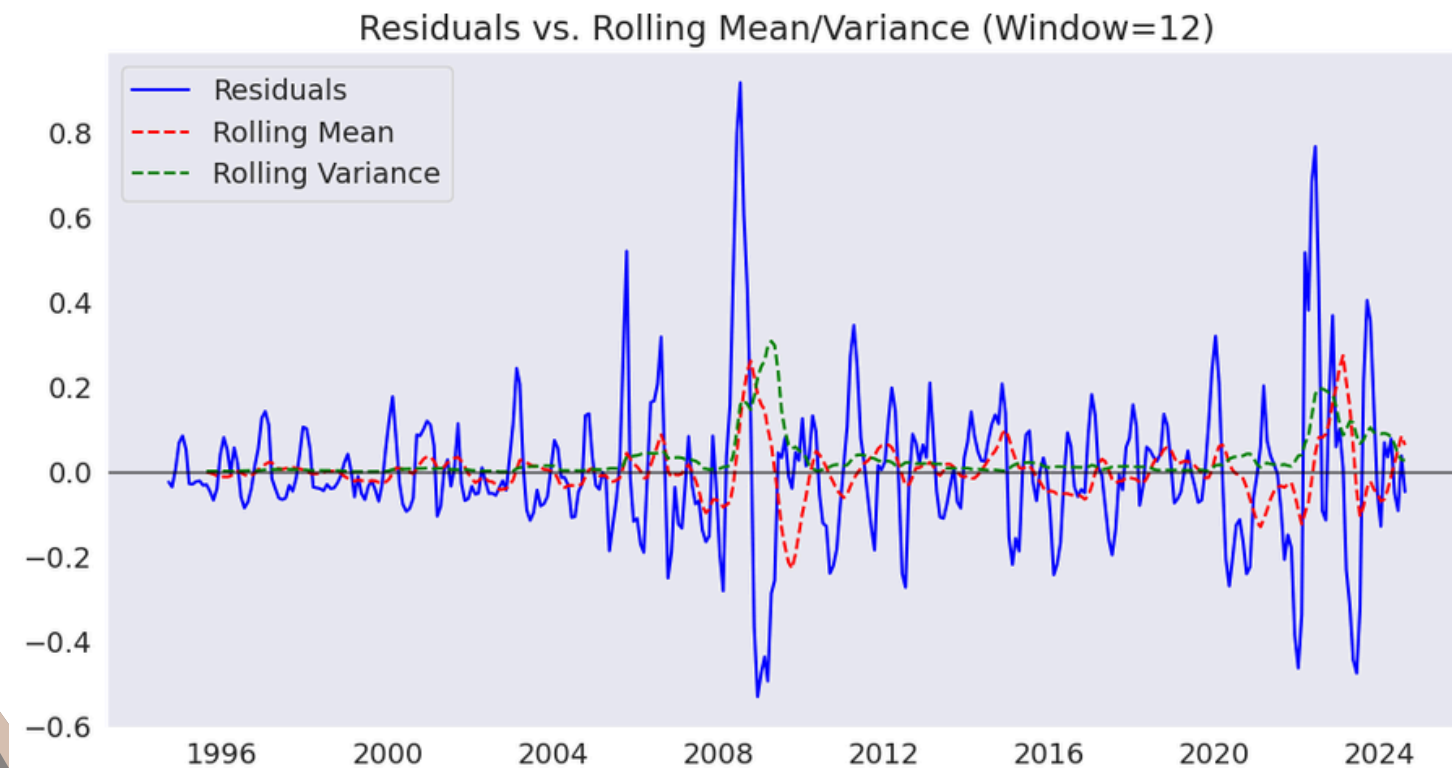
Railway Prices



- **Residuals:** Sharp variations are visible between 2002–2024, with increased deviations particularly after 2022.
- **Key Abnormality:** Post-COVID (2022): Relaxation of travel restrictions led to increased passenger demand, causing fare hikes.

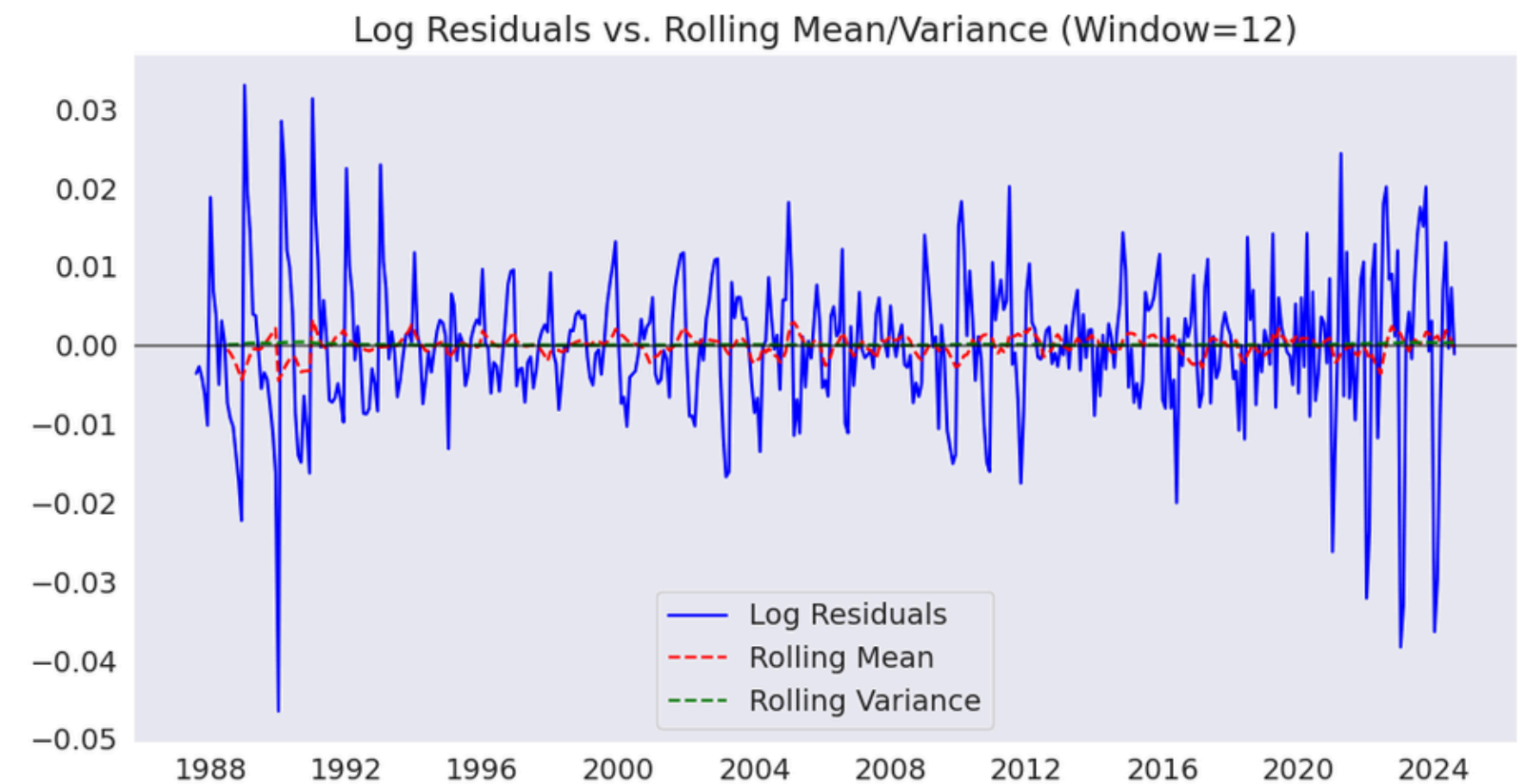
Stationarity Tests

Diesel Prices



```
ADF Test:  
p-value: 0.0000000015  
Conclusion: Stationary
```

Railway Prices

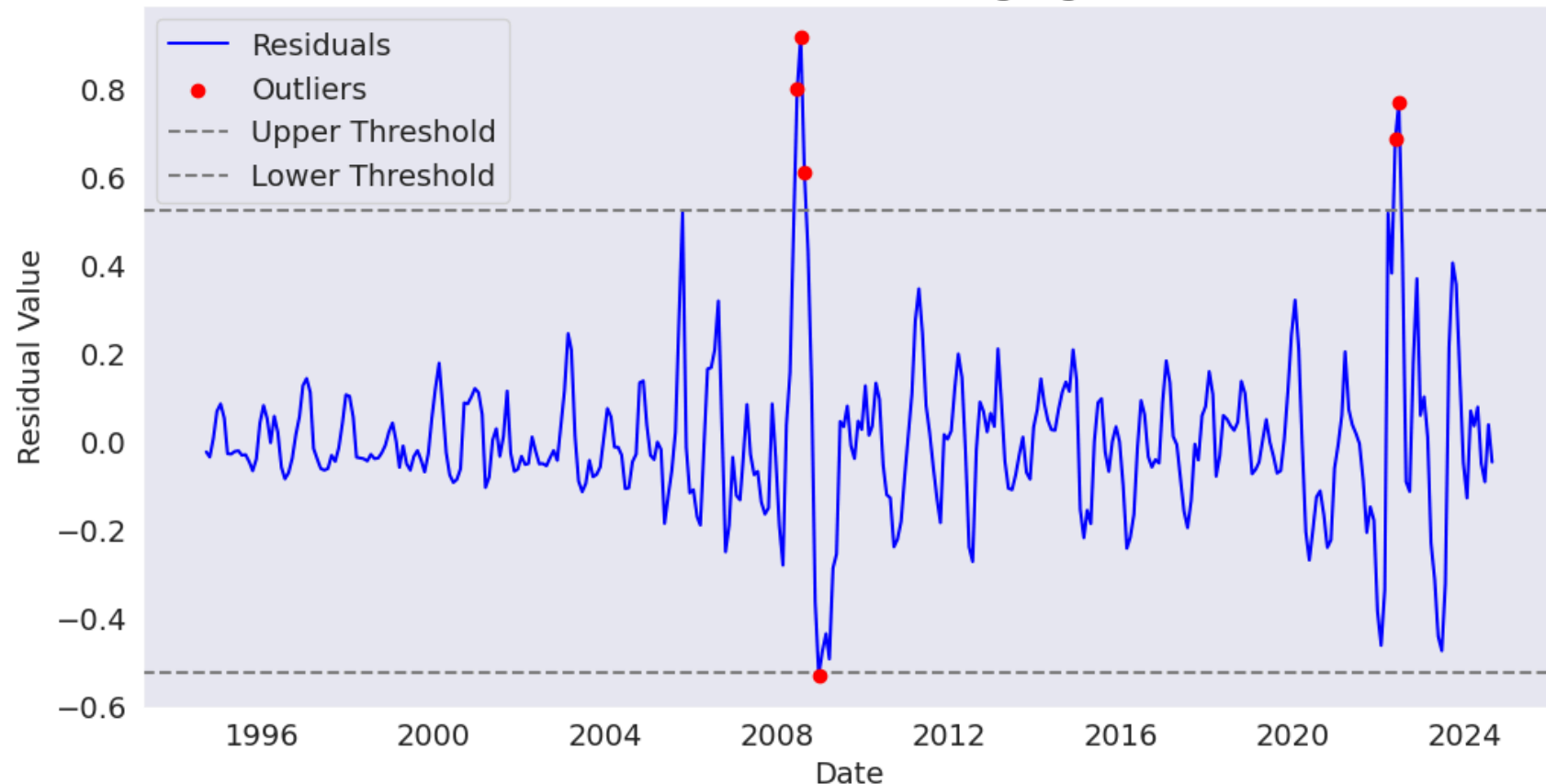


```
ADF Test:  
p-value: 0.000000001523  
Conclusion: Stationary
```

Outliers

Diesel Prices

Residuals with Outliers Highlighted



2007–2009 Spike:

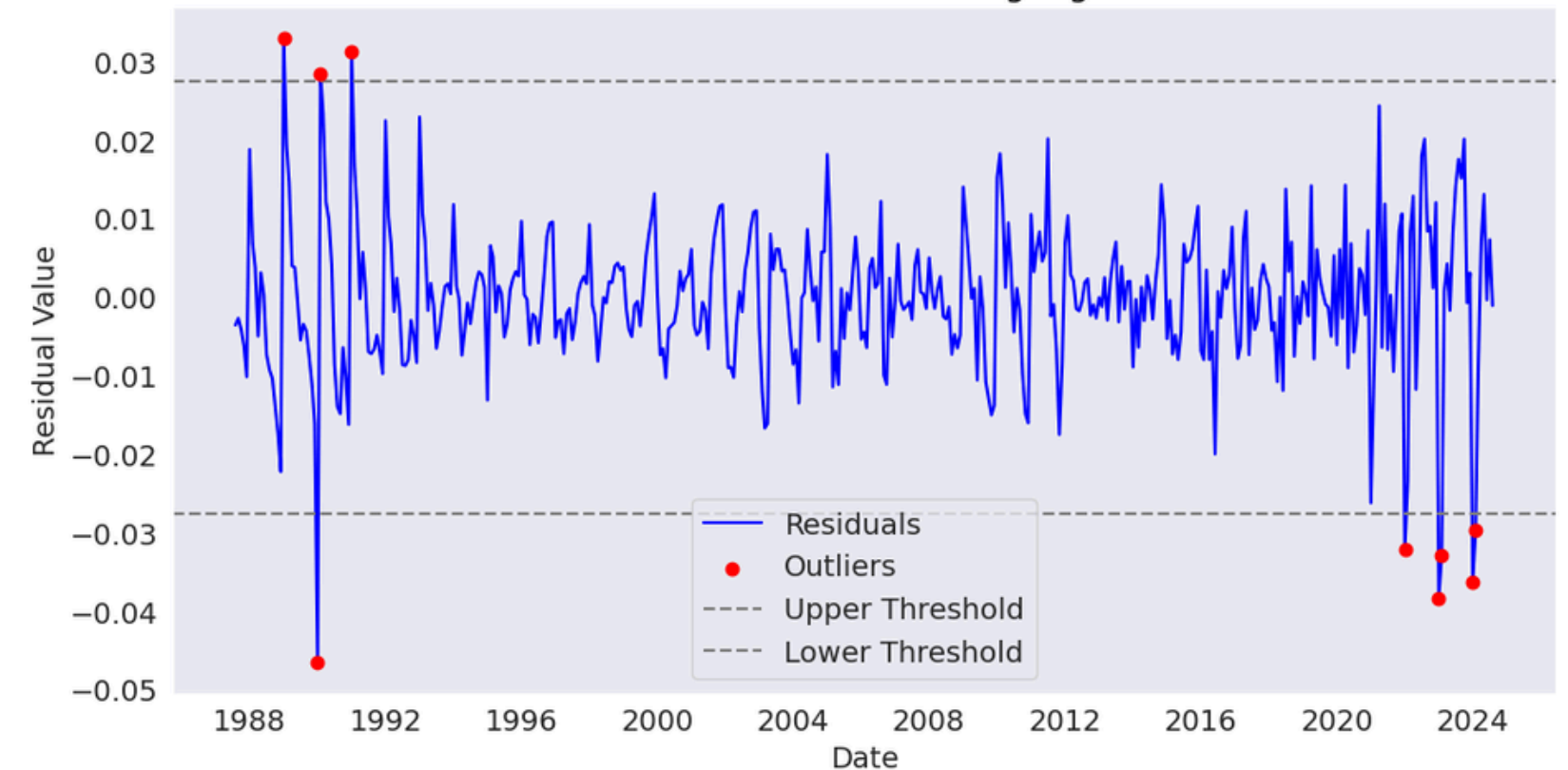
- **Global Financial Crisis:** Fuel prices initially soared in 2007, followed by a dramatic crash in 2008 due to recession and reduced global demand.

2022–2023 Spike:

- **Russia–Ukraine War:** Escalated geopolitical tensions led to oil supply disruptions and volatility in diesel pricing globally.

Railway Prices

Residuals with Outliers Highlighted

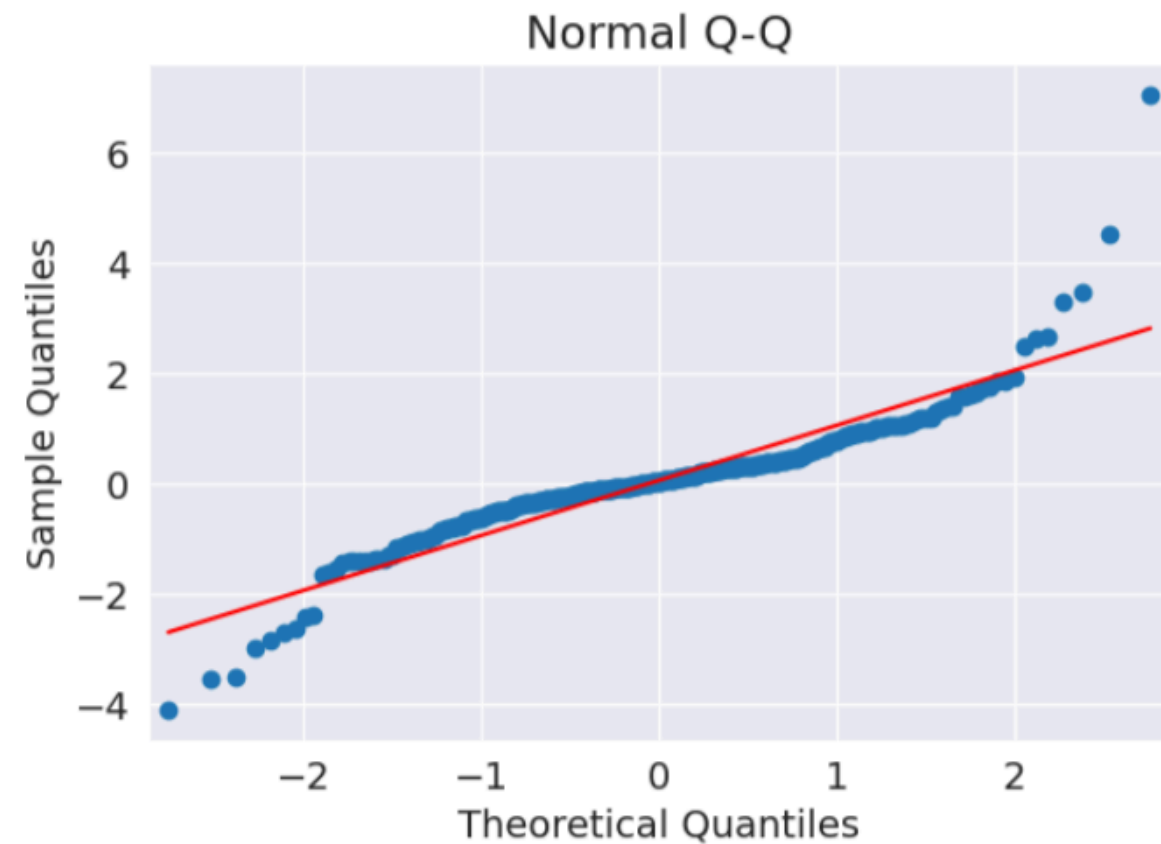


Outliers: 1988–1992 and 2022–2023

- **Late '80s–Early '90s:** Likely due to deregulation and pricing changes following the 1980 Staggers Rail Act, which allowed U.S. railroads more flexibility in setting fares.
- **2022–2023:** Post-COVID travel surge caused sudden spikes in demand, pushing prices up rapidly as restrictions were lifted.

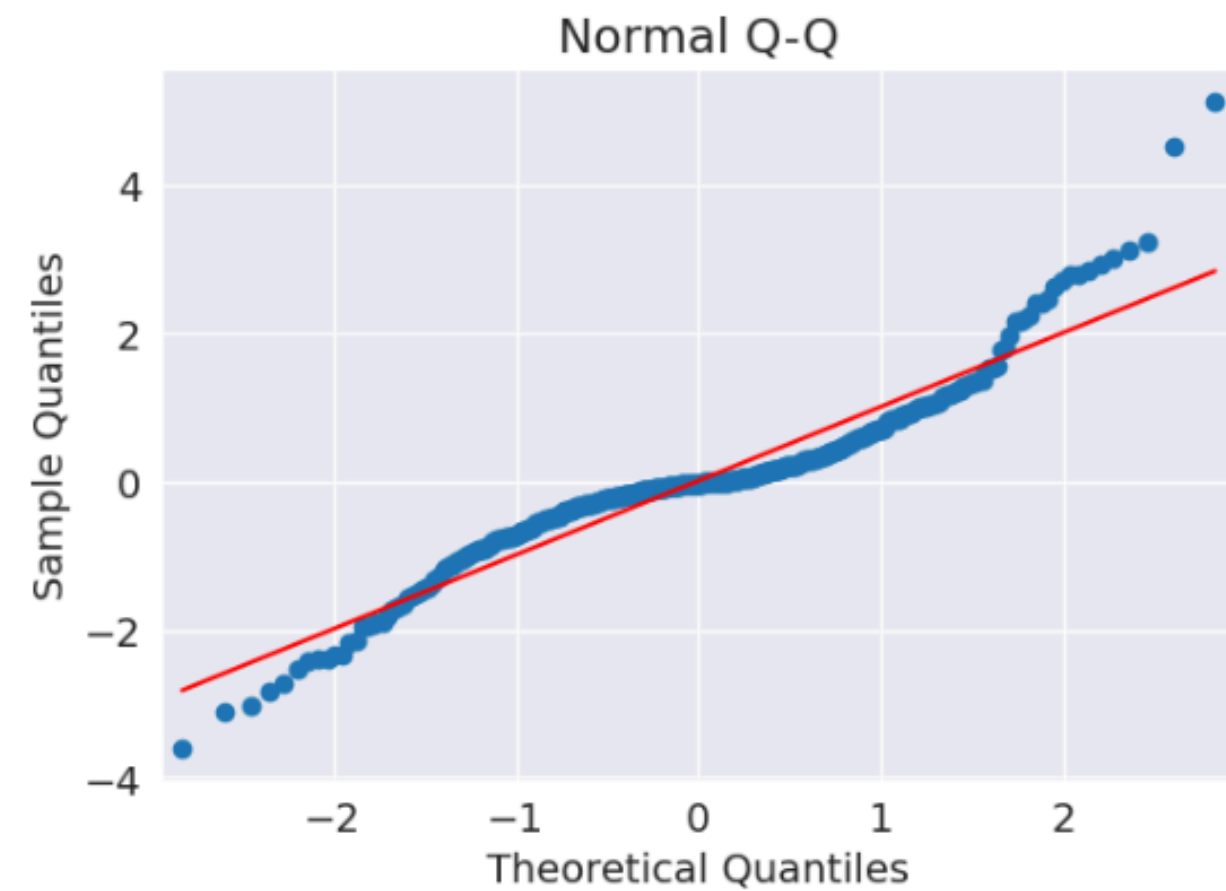
Quantiles

Diesel Prices



- Stronger deviation from normality than railway.
- More extreme outliers at both ends (below -4, above +6).
- Right tail longer → sudden price spikes (e.g., Russia-Ukraine war 2022).
- Left tail → sharp drops during 2008 recession or COVID demand crash.

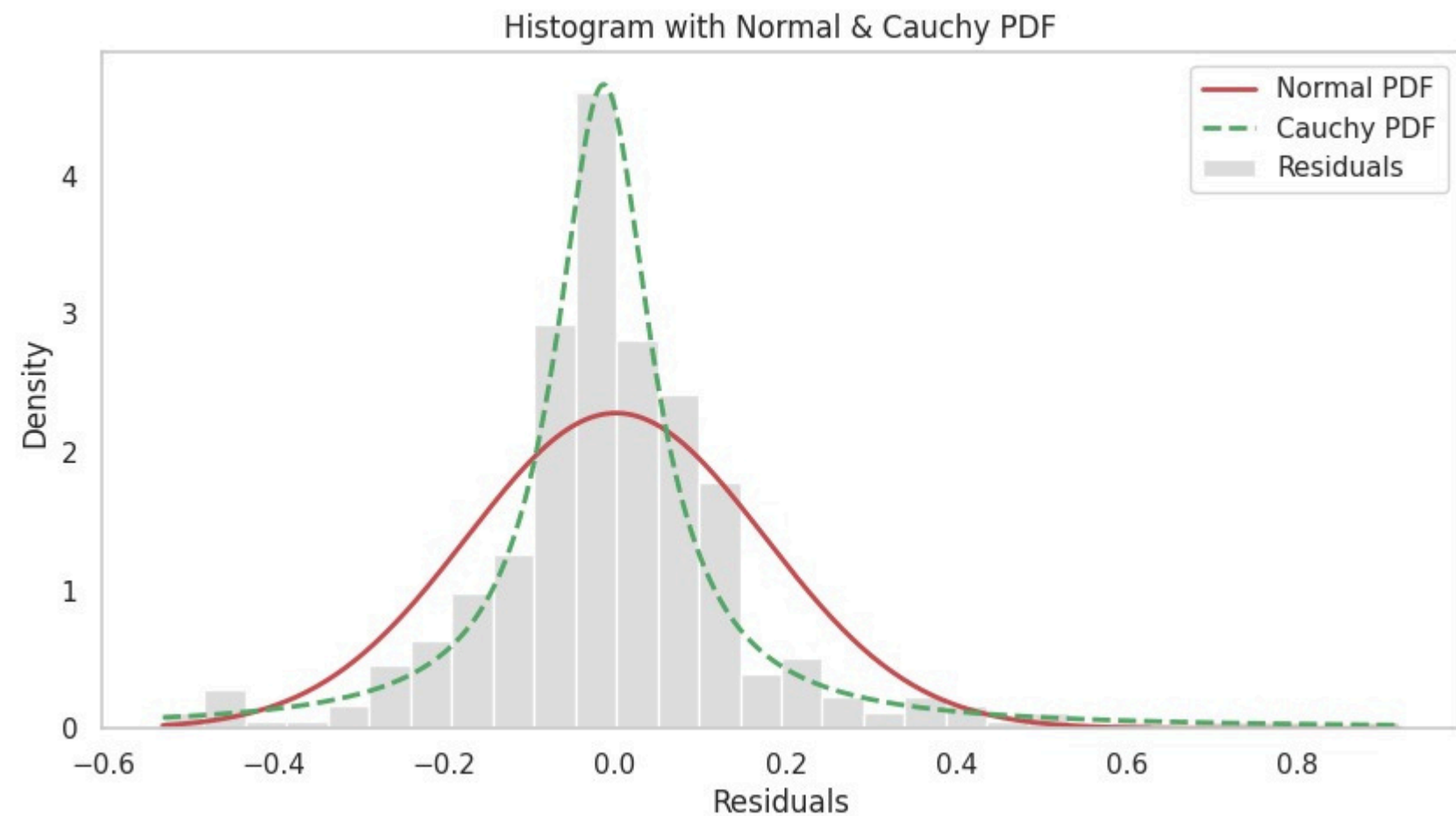
Railway Prices



- Shows heavy tails: data deviates from normality at extremes.
- Positive outliers above +4 suggest unusual fare increases.
- Negative outliers indicate rare dips, possibly during COVID-19 lockdowns.
- Overall, the middle data follows the line → moderate normal behavior.

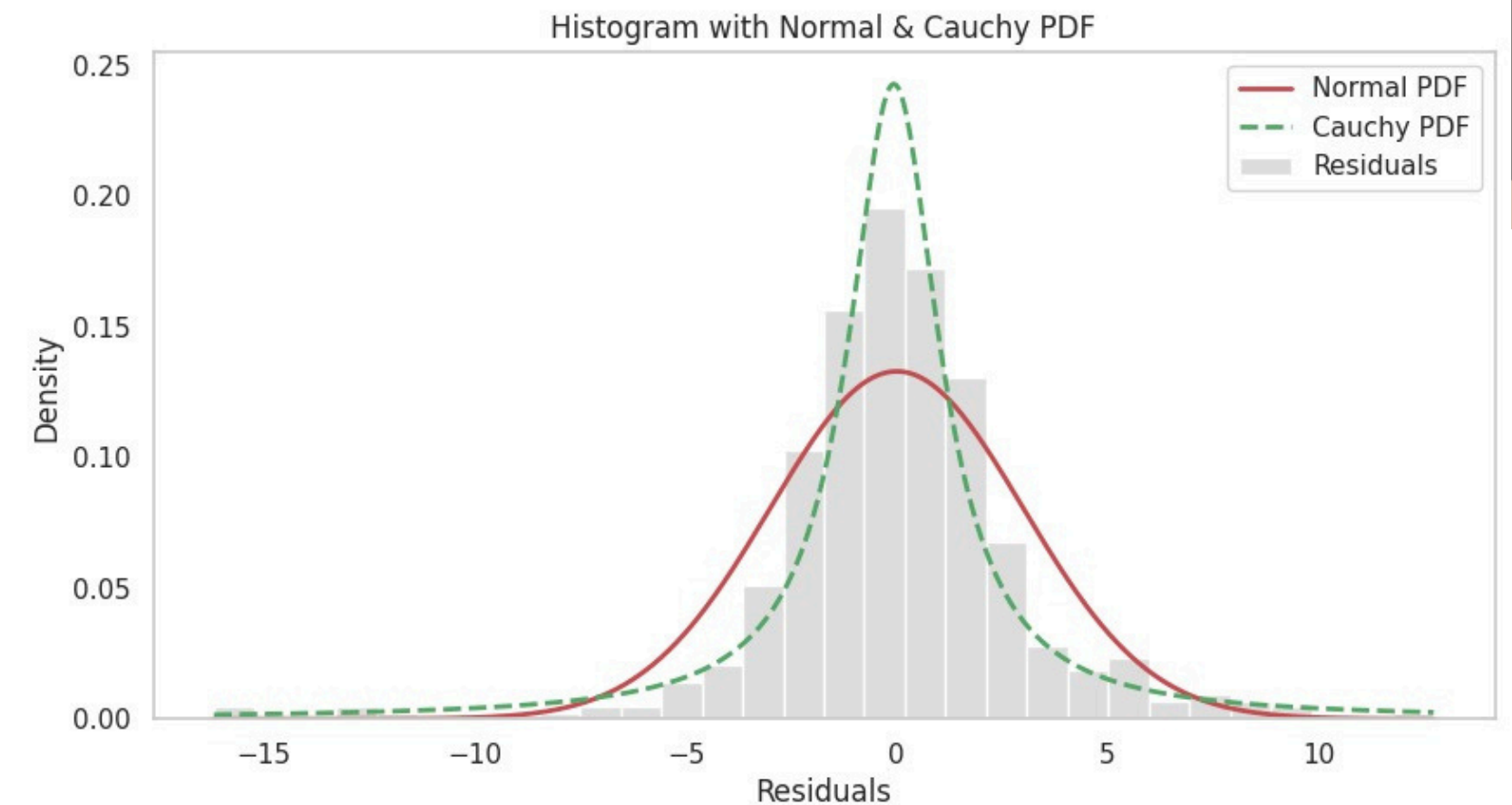
Distribution Fitting

Diesel Prices



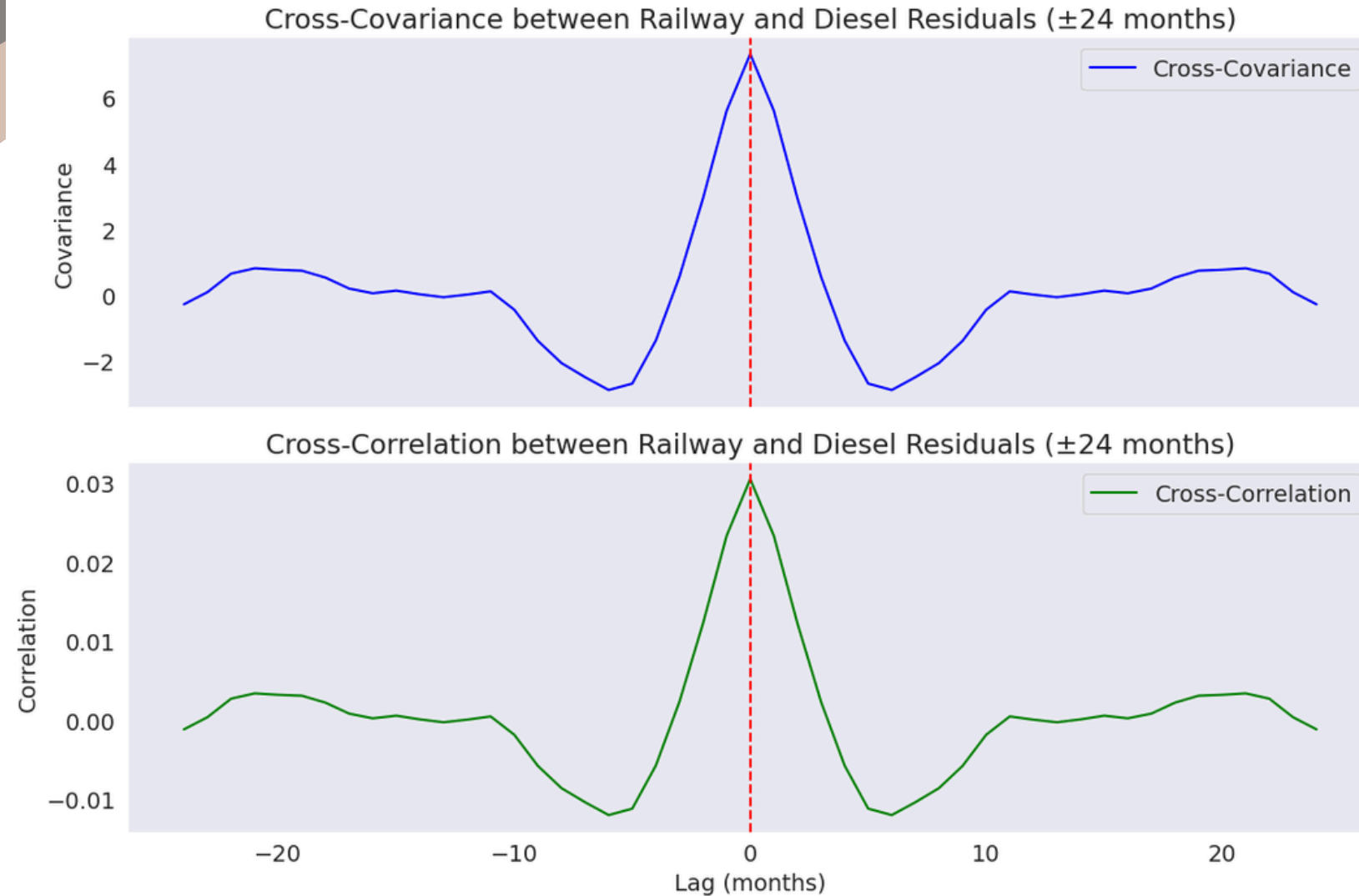
- Residuals deviate from the Normal curve and better align with the Cauchy distribution.
- Indicates presence of outliers or extreme values in pricing changes.
- Suggests non-Gaussian behavior, possibly due to irregular fluctuations in ticket pricing.

Railway Prices

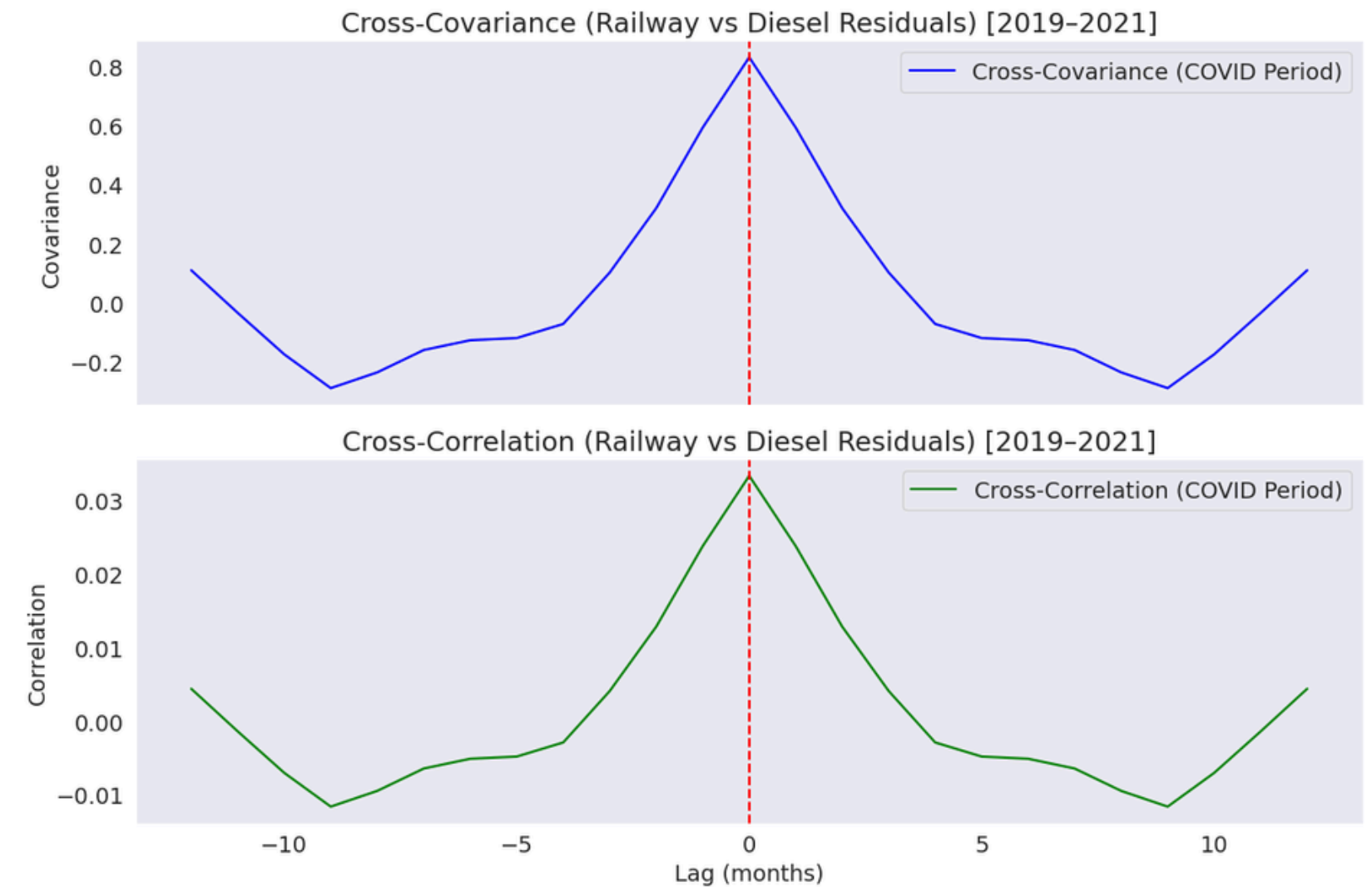


- Histogram showed a sharper peak and heavier tails, closely matching the Cauchy distribution.
- The normal curve (red) fits the center well but fails in the tails.
- The Cauchy distribution (green) captures the heavier tails better.

Cross - Correlation Analysis



- This analysis suggests that diesel prices affect railway prices almost immediately, as shown by the peak at lag = 0.
- There's limited evidence for longer lead/lag effects (e.g., prices reacting several months later).
- This could support policies or pricing strategies where railway fare changes are tightly coupled to current fuel costs.



- During the COVID-19 period, railway pricing were highly synchronized with diesel, reacting immediately (lag 0) to diesel price changes.
- **This could imply:** Pricing policies during COVID were tightly linked to fuel prices, there was less delay in policy decisions or fare updates in response to fuel cost changes, compared to more stable periods.

Conclusion

- We analyzed the relationship between US railway freight and diesel prices using residual-based correlation analysis (after removing trends and seasonality).
- Focused especially on the COVID-19 period to understand disruptions in correlation and behavioral patterns.
- Found that during COVID-19, the correlation weakened and became irregular, reflecting real-world shocks and instability.
- Over a longer random period, we observed periodic cross-correlation patterns, suggesting cyclical economic or seasonal links between the two.

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